Unit 1
Unit 1: HTML, CSS, Bootstrap & W3C — Practical/Oral Exam Notes

Context

Part of Web Application Development | Practical + Oral Exam Prep | SPPU Third Year IT

High-Yield Oral Topics

- **HTML5 Semantic Elements** — Examiners love this. Know 6+ examples.
- **CSS Flexbox vs Grid** — Always compared in viva.
- **Bootstrap Grid (col-md-6)** — Must explain responsiveness.
- **CSS Box Model** — Margin, Border, Padding, Content.
- **Inline vs Block vs Inline-Block** — Trap question.

MINI PROJECT ANCHOR: Student Management System (SMS)

> All examples in this unit use the **Student Management System** for consistency.
>
> - Frontend: HTML + CSS + Bootstrap
> - Features: Student Registration Form, Dashboard, Student List Table

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SECTION 1 — HTML

Exam Definition

HTML (HyperText Markup Language) is the **standard markup language** used to create the structure and content of web pages. It uses **tags** to define elements like headings, paragraphs, forms, and images.

Core Concept

HTML is NOT a programming language — it is a **markup language**. It tells the browser *what* to display, not *how* to behave. CSS handles appearance; JavaScript handles behavior.

Must-Know Tags (Viva Favorite)

Category	Tags	Purpose
Structure	`<html>`, `<head>`, `<body>`	Page skeleton

Category	Tags	Purpose
Headings	`<h1>` to `<h6>`	Hierarchy of text
Text	`<p>`, `<span>`, `<strong>`, `<em>`	Content
Links	`<a href="">`	Navigation
Lists	`<ul>`, `<ol>`, `<li>`	Ordered/Unordered
Tables	`<table>`, `<tr>`, `<th>`, `<td>`	Tabular data
Forms	`<form>`, `<input>`, `<button>`, `<select>`	User input
Media	`<img>`, `<video>`, `<audio>`	Rich media
Semantic	`<header>`, `<nav>`, `<section>`, `<article>`, `<footer>`	Meaningful structure

📌 HTML Forms and Structure

Instead of viewing full code, understand the theoretical composition of a form based on the syllabus:

- `<form>`: The container for input elements. The `action` attribute specifies where to send data, and `method` (GET/POST) specifies how.
- `<input>`: The most important form element. The `type` attribute determines if it's text, email, password, date, etc.
- `<select>`: Creates a drop-down list.
- `<label>`: Defines a label for a form element, improving accessibility. Use the `for` attribute.

🔥 Semantic HTML5 — MOST ASKED IN VIVA

Semantic Tag	Purpose	Non-Semantic Equivalent
`<header>`	Top section of page/section	`<div id="header">`
`<nav>`	Navigation links	`<div id="nav">`
`<main>`	Main content of page	`<div id="main">`
`<section>`	Thematic grouping of content	`<div>`
`<article>`	Self-contained content (blog post)	`<div>`
`<aside>`	Sidebar / related content	`<div id="sidebar">`
`<footer>`	Bottom of page	`<div id="footer">`
`<figure>`	Image with caption	`<div>`

> **Why Semantic HTML?** — SEO (Search Engine Optimization), Accessibility (screen readers), Code readability.

HTML Attributes — Viva Quick Fire

Attribute	Used On	Purpose
`href`	`<a>`	Link destination
`src`	`<img>`, `<script>`	File source
`alt`	`<img>`	Alternate text (accessibility)

Attribute	Used On	Purpose
`action`	`<form>`	Where to send form data
`method`	`<form>`	GET or POST
`type`	`<input>`	text, email, password, date, etc.
`required`	`<input>`	HTML5 validation
`id`	Any	Unique identifier
`class`	Any	CSS styling hook

🎤 VIVA Q&A — HTML

Q1: What is the difference between `<div>` and `<span>`?

> `<div>` is a **block-level** element — takes full width, starts on new line. Used for layout sections. `<span>` is an **inline** element — only as wide as its content. Used to style a portion of text.

Q2: What is the difference between GET and POST in forms?

> **GET** appends data to URL (visible, used for search). **POST** sends data in request body (hidden, secure, used for login/registration). POST is preferred for sensitive data like passwords.

Q3: What are HTML5 Semantic elements and why use them?

> HTML5 semantic elements like `<header>`, `<nav>`, `<article>` give **meaning** to the structure. Benefits: Better SEO (Google understands page structure), Accessibility (screen readers), and cleaner code.

Q4: What is the `DOCTYPE` declaration?

> `<!DOCTYPE html>` tells the browser to render the page in **HTML5 standards mode**. Without it, browsers go into "quirks mode" which causes inconsistent rendering.

Q5: What is `alt` attribute in `<img>`?

> The `alt` attribute provides **alternative text** that displays if the image fails to load. It is also used by **screen readers** for visually impaired users and improves **SEO**.

⚠️ Common Mistakes Students Make

- Forgetting `<!DOCTYPE html>` at top
- Not closing tags (especially `<li>`, `<td>`)
- Using `<b>` instead of `<strong>` (semantic vs presentational)
- Not adding `name` attribute to form inputs (data won't be sent)
- Mixing up `id` (unique) vs `class` (reusable)

🧠 Memory Trick

"Semantic HTML = You Know What's Inside Without Opening the Box"
`<article>` → article content. `<nav>` → navigation. `<footer>` → bottom info.

SECTION 2 — CSS

📌 Exam Definition

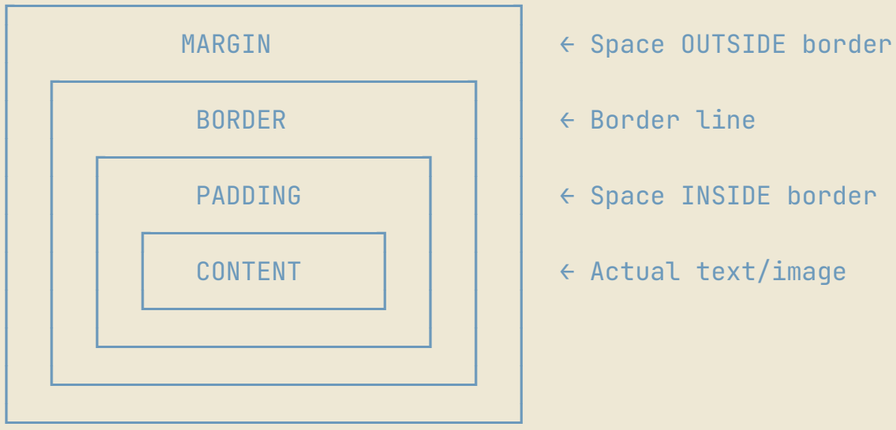
CSS (Cascading Style Sheets) is a **stylesheet language** used to control the presentation (layout, colors, fonts, spacing) of HTML elements. It **separates design from structure**.

🔑 Three Types of CSS

Type	Syntax	Priority	Use Case
Inline	`style="color:red"` inside tag	Highest	Quick one-off styles
Internal	`<style>` inside `<head>`	Medium	Single-page styles
External	`<link rel="stylesheet" href="style.css">`	Normal	Best practice, reusable

> **Cascade Rule:** Inline > Internal > External (when specificity is equal)

📦 The CSS Box Model — MUST KNOW



> **Viva Answer:** "Margin is OUTSIDE the border, Padding is INSIDE the border. Margin creates space between elements; padding creates space between content and its border."

📌 Core CSS Properties (Syllabus Highlights)

Instead of a long stylesheet, here are the essential properties you need to know for the exam:

- **Box Model Properties**: `margin` (outer space), `padding` (inner space), `border` (boundary), and `width`/`height`.
- **Positioning**: `static`, `relative`, `absolute`, `fixed`, `sticky`.
- **Layouts**:

- **Flexbox**: `display: flex;` for 1D alignment (rows or columns). Key properties: `justify-content`, `align-items`.
- **Grid**: `display: grid;` for 2D layouts. Key properties: `grid-template-columns`, `gap`.

- **Transitions & Animations**:
  - `transition: property duration timing-function;` for smooth changes.
  - `@keyframes` combined with the `animation` property for complex movements.

🔑 Flexbox vs Grid — MOST COMPARED IN VIVA

Feature	Flexbox	Grid
Dimension	1-Dimensional (row OR column)	2-Dimensional (rows AND columns)
Use Case	Navigation bars, card rows, centering	Page layout, dashboards
Direction	`flex-direction: row/column`	`grid-template-columns/rows`
Alignment	`justify-content`, `align-items`	`justify-items`, `align-items`
Example	Navigation links in a row	3-column dashboard

CSS Positioning — Viva Trap

Value	Behavior
`static`	Default. Normal flow. `top/left` ignored.
`relative`	Offset from its NORMAL position. Other elements not affected.
`absolute`	Removed from flow. Positioned relative to nearest positioned parent.
`fixed`	Removed from flow. Fixed to browser viewport (sticky navbar).
`sticky`	Normal flow until scroll threshold, then fixed.

🎤 VIVA Q&A — CSS

Q1: What is specificity in CSS?

> Specificity determines which CSS rule applies when multiple rules target the same element. Priority: Inline style (1000) > ID selector (100) > Class selector (10) > Element selector (1). Higher specificity wins.

Q2: What is the difference between `margin` and `padding`?

> Margin is the space **outside** the border (between elements). Padding is the space **inside** the border (between content and border). Background color fills the padding area but NOT the margin.

Q3: What is `box-sizing: border-box`?

> By default, width/height only includes content. With `border-box`, width/height includes padding and border too. This makes layout calculation much easier and predictable.

Q4: How does CSS animation work?

> CSS animation uses `@keyframes` to define style changes at different points, and `animation` property applies it to an element. Example: `animation: fadeIn 0.5s ease` — the element fades in over 0.5 seconds.

🧠 Memory Tricks

- **Box Model:** "My Big Precious Content" = Margin, Border, Padding, Content (outside to inside)
- **Flex is 1D, Grid is 2D** → Flex = one direction at a time, Grid = full table layout

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SECTION 3 — BOOTSTRAP

📌 Exam Definition

Bootstrap is a **free, open-source CSS framework** that provides pre-built responsive components and a **12-column grid system**, allowing developers to build mobile-responsive web pages rapidly without writing CSS from scratch.

Why Bootstrap over Raw CSS?

Aspect	Raw CSS	Bootstrap
Time	Write everything from scratch	Pre-built components ready
Responsiveness	Manual media queries	Built-in grid system
Browser Compatibility	Test/fix manually	Handled by Bootstrap
Consistency	Depends on developer	Consistent across team
Learning Curve	Deep CSS knowledge needed	Learn classes, apply fast

🔑 Bootstrap Grid System — MUST KNOW

Bootstrap divides every row into **12 columns**. You specify how many columns an element takes at each breakpoint.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Left Sidebar</div>
    <div class="col-md-8">Main Content</div>
  </div>
</div>
```

Bootstrap Breakpoints

Breakpoint	Code	Width	Target
Extra Small	`col-`	< 576px	Phones
Small	`col-sm-`	≥ 576px	Large phones

Breakpoint	Code	Width	Target
Medium	`col-md-`	≥ 768px	Tablets
Large	`col-lg-`	≥ 992px	Laptops
Extra Large	`col-xl-`	≥ 1200px	Desktops

> **Viva Tip:** `col-md-6` means "take 6 of 12 columns on medium screens and above" = **two equal columns** on tablet/desktop.

Key Bootstrap Classes — Cheatsheet

Class	Purpose
`container` / `container-fluid`	Centered wrapper / Full-width wrapper
`row`	Flex row (parent of columns)
`col-md-6`	Column taking 6/12 width on medium+
`btn btn-primary`	Blue button
`table table-striped`	Striped table
`card`, `card-body`, `card-title`	Card component
`navbar navbar-expand-lg`	Responsive navigation bar
`d-flex justify-content-center`	Flexbox centering
`mt-3`, `mb-4`, `px-2`	Margin/Padding utilities
`text-center`, `fw-bold`	Text alignment/weight
`d-none d-md-block`	Hide on mobile, show on desktop

🎤 VIVA Q&A — Bootstrap

Q1: What is the Bootstrap grid system?

> Bootstrap's grid system uses a series of containers, rows, and columns to layout content. It's based on a **12-column** flexible grid. You specify how many columns an element spans using classes like `col-md-6` (6 of 12 = half width on medium+ screens).

Q2: How does Bootstrap achieve responsiveness?

> Bootstrap uses **CSS media queries** internally. When you use `col-md-6 col-12`, Bootstrap applies `display: block; width: 100%` on small screens and `width: 50%` on medium+ screens automatically.

Q3: What is the difference between `container` and `container-fluid`?

> `container` has a fixed max-width that changes at breakpoints (centered with margins). `container-fluid` is always 100% wide — it spans the full viewport width.

Q4: Why use Bootstrap instead of writing CSS from scratch?

> Bootstrap saves development time with pre-built components (buttons, navbars, modals, tables), ensures cross-browser compatibility, and provides a proven responsive grid system. It's ideal for rapid prototyping and team projects.

⚠️ Oral Trap Questions

Trap: "Can you use Bootstrap AND custom CSS together?"

> Yes! You add Bootstrap via CDN, then write your own CSS file after it. Your custom styles override Bootstrap where needed. This is the standard approach.

Trap: "Bootstrap uses JavaScript — what for?"

> Bootstrap includes JavaScript (via Popper.js) for **interactive components** like modals, dropdowns, carousels, tooltips, and collapsible navbars. Pure CSS cannot handle these interactions.

SECTION 4 — W3C

📌 Exam Definition

W3C (World Wide Web Consortium) is the **international standards organization** that develops and maintains open web standards to ensure long-term growth and interoperability of the web.

Key Points for Viva

Aspect	Details
Founded by	Tim Berners-Lee (inventor of WWW) in 1994
Purpose	Set standards for HTML, CSS, XML, Web Accessibility
Key Standards	HTML5, CSS3, WCAG (Accessibility), SVG, WebRTC
Why Important	Ensures all browsers render the same code consistently

🎤 VIVA Q&A — W3C

Q1: What is W3C and why is it important?

> W3C (World Wide Web Consortium) is the body that creates and maintains web standards. Without W3C, each browser would implement HTML and CSS differently, breaking cross-browser compatibility. W3C ensures that code written by one developer works consistently everywhere.

Q2: What is WCAG?

> WCAG (Web Content Accessibility Guidelines) is a W3C standard that defines how to make web content accessible to people with disabilities (visual, auditory, cognitive). It defines criteria like sufficient color contrast and alt text for images.

📋 PRACTICAL FILE EXPECTATIONS — Assignment 1

Your practical file for Assignment 1 should show:

1. **HTML Registration Form** — Student registration with name, email, course, DOB fields

2. **Bootstrap Dashboard** — Sidebar + 3 statistics cards + student table
3. **CSS Styling** — External stylesheet with responsive design
4. **JavaScript Validation** — Form validation + store data in `localStorage`
5. **AJAX POST** — Submit form data via AJAX and display success message
6. **Student List Page** — Read from localStorage and display in table

Viva Flow for Assignment 1:

> "I created a Student Management System frontend. The registration page uses **Bootstrap grid** (col-md-3 for sidebar, col-md-9 for content). The form uses **HTML5 validation** (required, type="email"). On submit, **JavaScript** validates the data and stores it in **localStorage** as a JSON array. I also used **AJAX POST** to send the data to the backend. The dashboard shows student statistics using **Bootstrap cards**."

🔥 High Probability Exam Questions

1. What is the difference between HTML and HTML5? (Answer: HTML5 added semantic elements, audio/video support, canvas, local storage, form validation)
2. Explain the CSS Box Model with diagram
3. What is Flexbox? How is it different from Grid?
4. What is Bootstrap Grid? Explain with `col-md-6` example
5. What are semantic elements? Give 5 examples
6. Difference between `margin` and `padding`
7. What is media query? How does it work?
8. What is specificity in CSS?
9. What is `DOCTYPE`?
10. What is the difference between `id` and `class` in CSS?

🧠 Quick Revision Table

Topic	Key Point	Viva One-Liner
HTML	Markup language for structure	"HTML tells browser WHAT to show"
CSS	Stylesheet for presentation	"CSS tells browser HOW to show it"
Bootstrap	CSS framework with 12-col grid	"Responsive design without writing media queries"
Semantic HTML	Tags with meaning	"Screen readers and SEO benefit"
Flexbox	1D layout	"Row OR column alignment"
Grid	2D layout	"Rows AND columns layout"
Box Model	M-B-P-C	"Margin outside, Padding inside border"
W3C	Standards body	"Tim Berners-Lee's web standards org"